CAUSAL ML · UPLIFT MODELING

The best model wasn't the fanciest one.

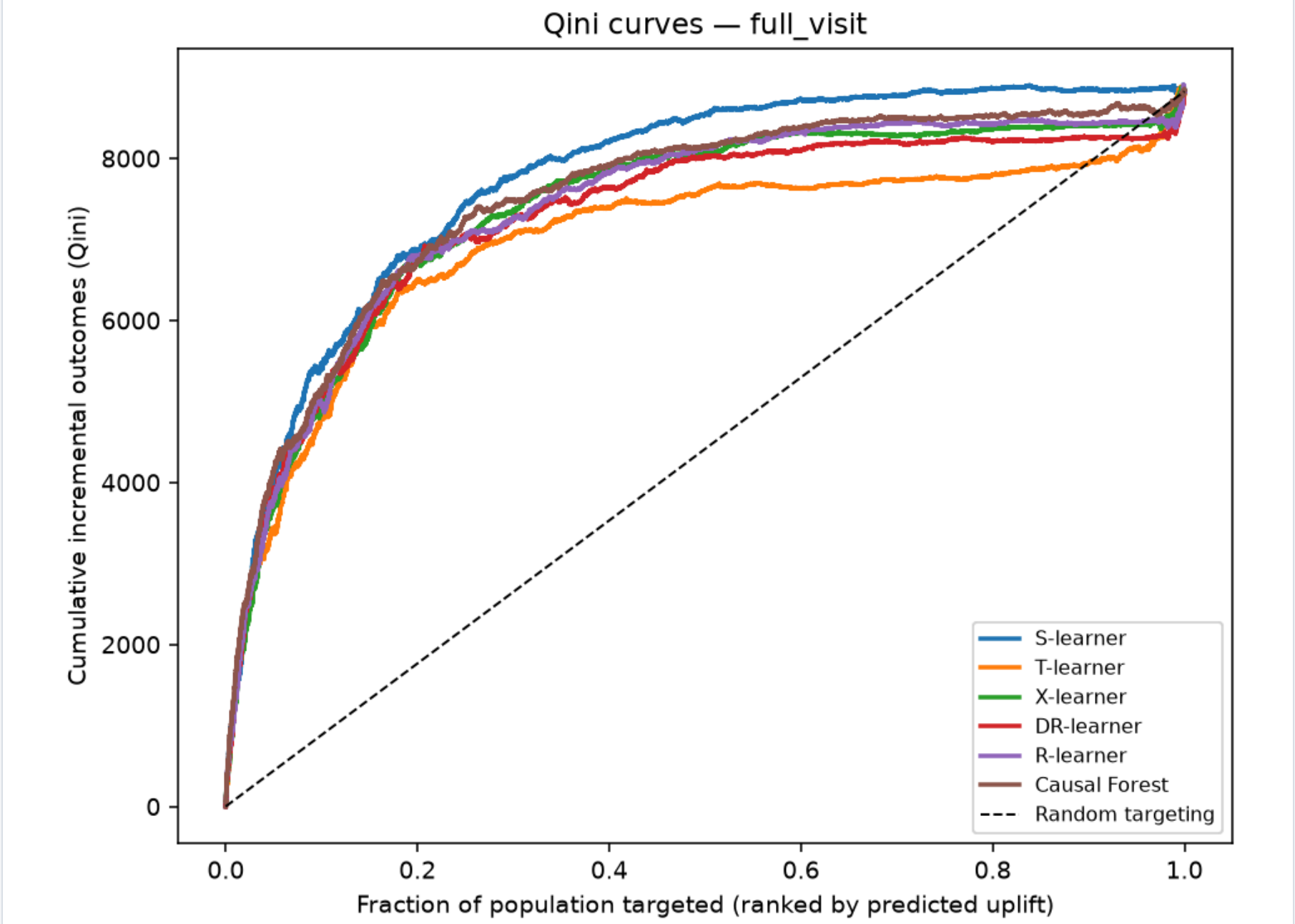
Benchmarking six causal-ML estimators on 14 million randomized ad exposures.

swipe →

Bruno Karoubi

01 · TARGETING

You don't need to treat everyone.



The top **10%** of users by predicted uplift carry **~82%** of the incremental visits. Half the audience is dead weight — and invisible if you only look at the average effect.

02 · ESTIMATOR CHOICE

Same data. Six methods. Different answers.

SPREAD IN TARGETING QUALITY (QINI, @500K)

2.6×

between the best and worst estimator

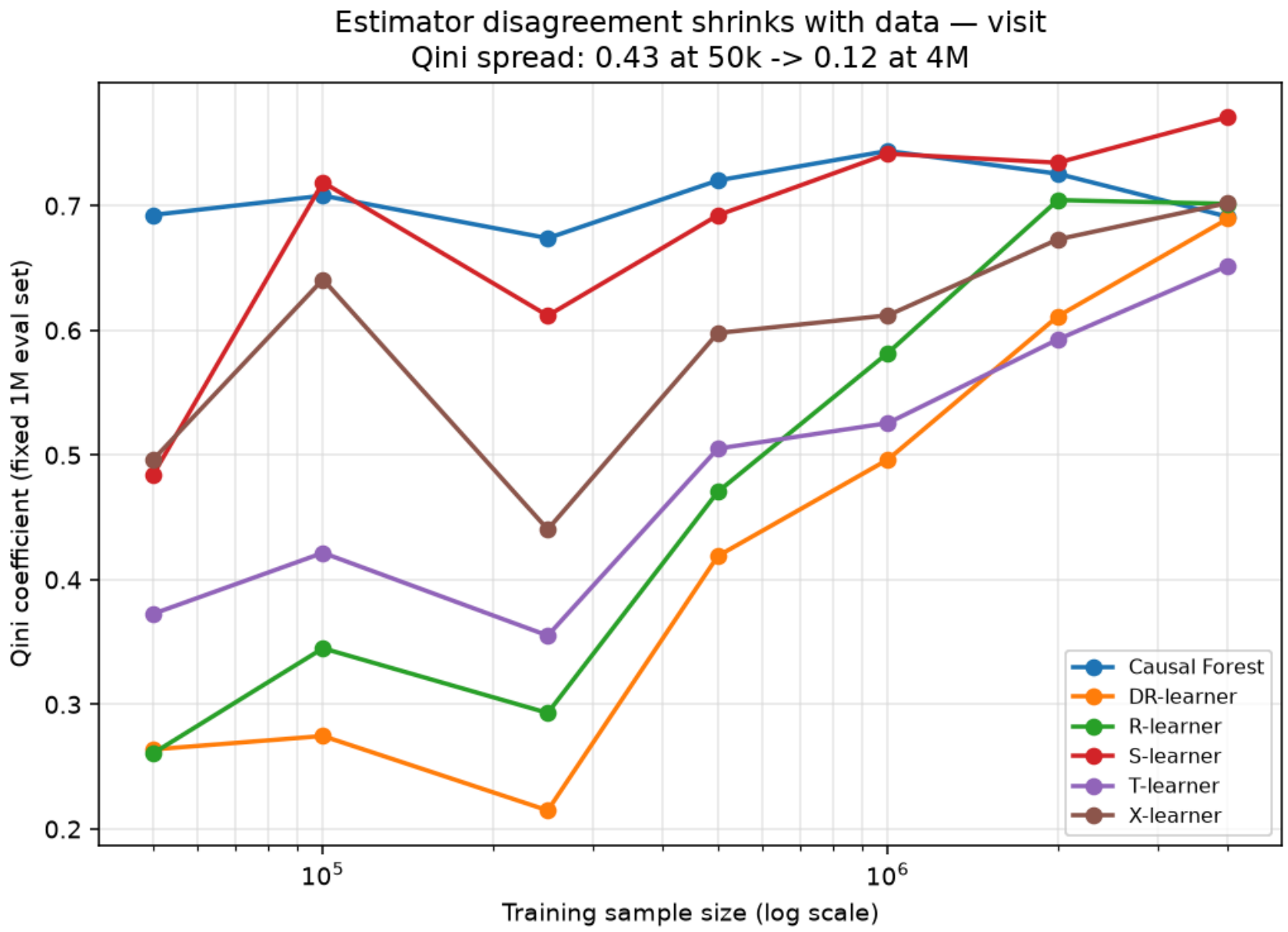
USERS FLAGGED AS ACTIVELY "HARMED"

4% to **44%** — on identical data, depending only on which estimator you picked.

S / T / X-learner · DR · R-learner · causal forest

03 · THE TWIST

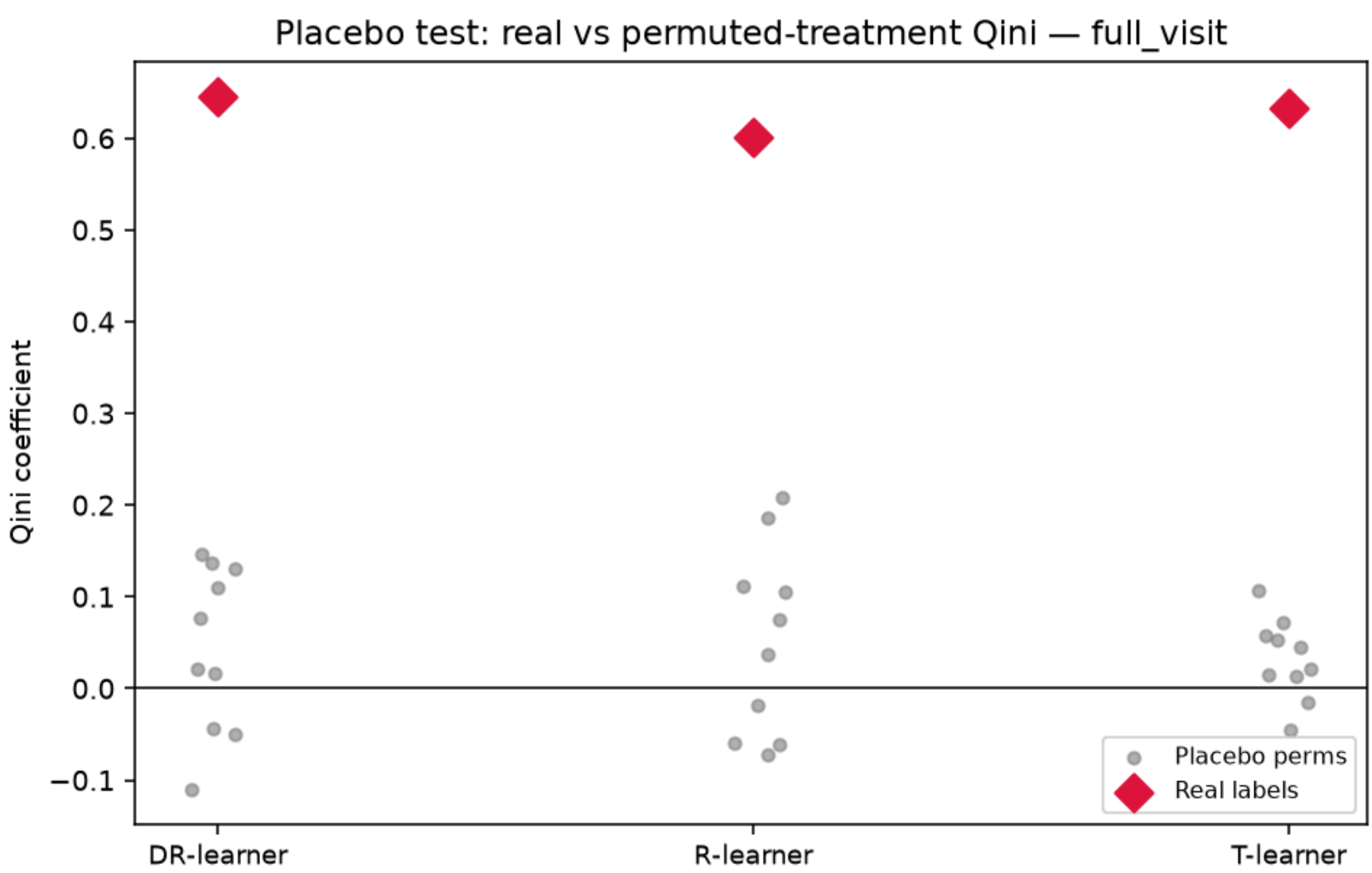
The disagreement was a sample-size illusion.



From 500k to 4M rows the estimators converge — the gap between best and worst shrinks **0.43 → 0.12**. Estimator choice matters most exactly when you have the **least** data.

04 · THE RIGOR CHECK

Shuffle the treatment. A real model finds nothing.



The DR-learner vs. its own placebo null: **z = 1.4** at 500k (fails) → **z = 7.0** at 4M (passes). It was never overfitting — it was **under-powered**.

TAKEAWAYS

- Targeting, not averages, is the business case.
- The regularized methods (S-learner, causal forest) were robust at every scale.
- The placebo test is the discipline that tells overfitting from under-power.

Full write-up + reproducible code → **link in the comments.**

14M-row Criteo uplift benchmark

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